

Predictor and control measurements

Advanced Scenario-Based Question

A large software company, **TechNova Solutions**, is developing a **cloud-based hospital management system** intended for deployment across multiple countries. The project involves distributed teams working in different time zones and follows an **Agile-Scrum methodology**.

During the 5th sprint review, the senior management team observes several concerning trends and requests a detailed metrics-based analysis before deciding whether to:

- Continue with the current development approach
- Increase resources
- Refactor parts of the system
- Delay the release

The following data has been collected over the last 3 sprints:

Collected Measurements

1. Average developer effort increased from **120 person-hours/week to 190 person-hours/week**
2. Cost per sprint increased by **35%**
3. Number of code commits per developer decreased significantly
4. Cyclomatic complexity of core modules increased from **8 to 22**
5. Defect density in testing phase increased from **0.5 defects/KLOC to 1.8 defects/KLOC**
6. Average time to resolve defects increased from **2 days to 6 days**
7. Code review rejection rate increased from **10% to 38%**
8. Test case execution coverage dropped from **85% to 60%**
9. Customer-reported usability issues increased during beta testing
10. Build failure rate increased from **5% to 25%**
11. Team velocity (story points completed) decreased from **45 to 28**
12. Number of change requests from stakeholders increased significantly

Additional Context

- Management is unsure whether the problem lies in **process inefficiencies** or **product quality degradation**
- Some managers argue that increasing **effort and cost** indicates a process issue
- Others argue that **code complexity and defect trends** indicate future product risks
- A proposal suggests halting development temporarily for **code refactoring**
- Another proposal suggests increasing team size to speed up delivery

Question Tasks

Part A (Identification)

From the above measurements:

1. Classify each metric as either:
 - **Control Metric**
 - **Predictor Metric**
2. Be careful: some metrics may appear ambiguous. You must justify your classification.

Metric	Type	Reason
Effort (person-hours)		
Cost per sprint		
Code commits per developer		
Cyclomatic complexity		
Defect density		
Defect resolution time		
Code review rejection rate		
Test coverage		
Customer usability issues		
Build failure rate		
Team velocity		
Change requests		

Part B (Critical Reasoning – Core of Question)

Explain in detail:

- Why each selected metric belongs to its category
- What makes certain metrics **confusing or borderline cases**
- How misclassification could lead to **wrong management decisions**

Part C (Decision Justification – Advanced Thinking)

Management wants to take action based on these metrics.

1. Which **3 Control Metrics** should management prioritize immediately?
2. Which **3 Predictor Metrics** indicate serious future risks?
3. Should the company:
 - Increase resources
 - Refactor code
 - Continue as is
 - Delay release

☞ Provide a **well-justified recommendation** based on the interaction of both metric types.